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ORIGINAL ARTICLE

Developing digital transformation leadership for digital-age organizational agility: a pragmatic framework

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Abstract
Digital transformation is a complex and multi-tiered process of organizational change aimed at sustaining digital-age organizational agility and requiring continuous effort. To help organizations better respond to the increasingly dynamic and disruptive nature of digital environments, this paper presents a unique pragmatic framework of digital transformation leadership developed in over a decade of digital transformation research and practice. The framework builds on and articulates the interdependent relationship between digital transformation, agility, and leadership, emphasizing the role of individual and collective practices in transforming an organization. The framework comprises a theoretical account of digital transformation and digital transformation leadership, a model of digital transformation leadership constituting a collection of practical tools for initiating and guiding digital transformation efforts, and a process logic for using the framework in organizations. This paper contributes to theory by extending the leadership-as-practice agenda into the digital transformation domain and addresses a critical practical gap by offering a generative tool allowing practitioners to design and maintain their digital transformation in an inclusive and context-sensitive way.

Keywords Digital transformation · Organizational agility · Leadership-as-practice · Organizational sensemaking · Paradox management

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1 Introduction

Recent efforts conceptualizing digital transformation take it to be divergent from other technology-enabled organizational transformations (Hanelt et al. 2021; Korhonen 2023; Omol 2024). Digital transformation requires organizations to continuously adjust their ways of working rather than pursue a one-time episodic change (Hanelt et al. 2021). That is due to the fact that digitally disruptive competitive environments make organizational agility a prerequisite for organizations' survival (Verhoef et al. 2021; Vial 2019). All the while organizational agility is an ongoing process (Harraf et al. 2015), especially as the progress of digital technologies results in continuous and rapid changes to organizations' competitive environments (Vial 2019). Approaching organizational agility in the digital age, therefore, is a continuous process of strategic organizational change.

Taken as a means to remain agile in the digital age, digital transformation cannot be understood as a technological or IT-enabled organizational change only (Baiyere et al. 2020; Korhonen 2023). With digital transformation constituting an organizational response to organizations' changing environments, there is a further need to elaborate on the interdependent relationship between digital transformation and organizational agility (Ciampi et al. 2022), especially when it comes to understanding the practice of organizational agility (Girod et al. 2023). Approached as a continuous organizational phenomenon, digital transformation requires organizations to solve complex multi-tiered challenges. Digitally transforming organizations must engage with and manage paradoxical tensions inherent to organizing (Andriopoulos and Lewis 2009; Danneels and Viaene 2022; Soh et al. 2023). At the same time, organizations still struggle with identifying their digital transformation needs (Baiyere et al. 2025). Digital transformations require cultural change (Butt et al. 2024), new approaches to management and leadership, and new models of collaboration (Hess et al. 2016; Korhonen 2023; Weber et al. 2022). However, despite employee resistance still constituting a prominent reason for digital transformation failure (Cieslak and Valor 2025), digital transformation research only starts to explicitly acknowledge the role of individual workers in figuring out new modes of human-technology interactions and scaling new digitally-driven ways of working (Ashrafi et al. 2025).

The effort of individuals in driving the transformation is complicated by it remaining under the radar of digital transformation practice and research (Ashrafi et al. 2025), as well as by the knowledge embedded in changing work practices being tacit rather than explicit (Schümmer and Haake 2009). To remain agile in light of the relentless progress of digital technologies, digital transformation needs to be continuously created by organization members. To be maintained over time, digital transformation needs to rely on and acknowledge the individual and collective efforts supported by mechanisms and tools grounded in the understanding of organizational change in the digital age. This paper aims to build on this realization and to recognize that digital transformation requires acknowledging, making explicit, and orchestrating the efforts of developing and scaling new digitally-driven ways of working. Digital transformation, then, arises to be a complex, multi-tiered, and emergent endeavor that requires appropriate tools and a theoretical base.

Creating knowledge on digital transformation understood as a multi-tiered process of organizational change, requires a different theoretical grounding. In recent years, pragmatic philosophy has been revisited by organizational scholars for dealing with complex and systemic organizational phenomena (Farjoun et al. 2015). That is due to the fact that pragmatism does not simplify the dynamic nature of the world or the ongoing flow of experience in which agents find themselves. Therefore, it supports engaging with temporal and relational aspects of organizations within the frameworks of emergence and creation of meaning (Helin et al. 2014; Langley and Tsoukas 2016; Simpson and den Hond 2022). In this way, pragmatism supports creating for-practice "policy-making" knowledge for organizations, while maintaining a more complex and dynamic interpretation of the world necessary for engaging with the convoluted subject of organizational change (Farjoun et al. 2015).

Pursuing a similar ambition of creating practical knowledge on digital transformation as ongoing organizational change towards digital-age organizational agility, we put forward a pragmatic framework for developing digital transformation leadership. This framework was developed in response to a research question reflecting a practical need of digitally transforming organizations: *How can organizations achieve digital-age organizational agility by developing digital transformation leadership?* The framework has been developed and put to use in a multitude of digitally transforming organizations between 2014 and 2024, while a variety of their experiences with initiating and maintaining digital transformation have been examined. It aims to, with appropriate guidance, allow practitioners to approach digital transformation of their organization as organizational change that they can design and maintain in a reflected way. Using the framework would mean organizing for inclusive and holistic continuous digital transformation, while creating and sharing practical knowledge in the form of new digitally-driven ways of working appropriate for their organization. This collective action of organization members engaging in the process of pragmatic inquiry to establish new ways of working with digital technologies for the purpose of digital-age agility is what is referred to in this paper as digital transformation leadership.

The presented framework comprises three parts, collectively covering five design components (A to E):

(1) A theoretical account approaching digital transformation leadership from a practice perspective, as a process of maintaining continuous digital transformation towards digital-age organizational agility, summarized in analytical propositions (*Component* A).
(2) A model of digital transformation leadership, consisting of three practical tools to assist organizations in highlighting, supporting, and coordinating processes of digital transformation leadership. The model consists of (1) a 2 × 2 canvas of digital transformation paradoxical tensions forming four digital transformation practice areas – "vigilant," "voyager," "visionary," and "vested" (*Component* B); (2) a list of "jobs to be done" outlining lower-level digital transformation progress points (*Component* C), and (3) a list of "digital transformation leadership qualities" capturing the behavioral qualities that contribute to the success of digital transformation practices in each practice area (*Component* D).

(3) A proposed process logic describing a designated way of using the model for initiating and maintaining digital transformation leadership across an organization (*Component* E).

The framework pushes the boundaries of the practice of and the scholarship on digital transformation. Theoretically, by approaching digital transformation, organizational agility, and leadership from the practice perspective, the framework contributes to the leadership-as-practice (Carroll et al. 2008; Raelin 2023) agenda, extending it to the field of digital transformation. Additionally, it answers the call for theoretical developments in the research field of digital transformation, which is dominated by case studies whose insights are limited in space and time (Packmohr et al. 2020). It also connects the concepts of digital transformation, organizational agility, and leadership. Thereby, it explicates the interdependent relationship between digital transformation and organizational agility and opens the door for realizing and leveraging leadership important for digital transformation. Building upon over ten years of gathering epistemic (*theoria*) and practical (*praxis*) knowledge on digital transformation, the framework is developed as a generative (*poietic*) design artifact, allowing practitioners to creatively apply and adopt its guidance to the context of their organizations.

Practically, the framework provides organizations with a common language and reference frame around digital transformation and its leadership. It outlines four digital transformation practice areas to be led, cultivated, and balanced for organizations to develop and maintain digital-age organizational agility – vigilant, voyager, visionary, and vested. It provides lower-level guidance by defining the progress that digitally transforming organizations and engaged practitioners need to strive for in each practice area. Additionally, it identifies digital transformation leadership qualities – behavioral qualities showing up most vividly in each leadership context, contributing to the success of new practices, and shaping the environment in which they were introduced. Thereby, it calls for acknowledging and cultivating the diversity and contingent nature of human expression that can facilitate the development of digital transformation leadership. Finally, we provide recommendations for how the framework should be put to work in organizations in a way that acknowledges their organizational context and starting point.

This article is organized as follows. Given that the presented pragmatic framework is grounded in a novel understanding of digital transformation, we omit the discussion of related works as no works follow a similar ambition. Instead, the literature on digital transformation, organizational agility, and leadership is synthesized into a theoretical account of digital transformation leadership (Sect. 3). Therefore, Sect. 2 proceeds to outline the development of the framework introduced in this article. Section 3 provides a theoretical account of digital transformation leadership, articulated through propositions that make sense of the connection between digital transformation, organizational agility, and leadership. Section 4 then presents a model of digital transformation leadership synthesized from the presented theoretical account. The model comprises a 2×2 canvas of digital transformation paradoxical tensions forming four digital transformation practice areas (*Component* B; Sect. 4.1), as well as a list of progress points (jobs to be done) (*Component* C; Sect. 4.2) and digital transformation leadership qualities associated with each practice area (*Component*

D; Sect. 4.3). The final component of the framework (*Component* E) is presented in Sect. 5 – a process logic utilizing paradox management to operationalize the model and guide cohesive, context-sensitive digital transformation efforts. Section 6 discusses the main theoretical and practical contributions of the work presented in this article, highlights its limitations, and suggests directions for future research.

2 Methodology

The framework was created in response to the research question of *how organizations can achieve digital-age organizational agility by developing digital transformation leadership.* Aiming to produce prescriptive framing knowledge that is actionable and theoretically grounded, we worked within the pragmatic research tradition (Ågerfalk 2010; Farjoun et al. 2015; Kelly and Cordeiro 2020) following an abductive theory-building process (van Hulst and Visser 2025). Following an abductive process is suitable for pragmatic "policymaking-oriented" knowledge creation, as it allows for leaving room for both theoretical insight as well as surprises and tensions that emerge between, for example, "policies on paper" and actual "policymaking" on the ground (van Hulst and Visser 2025). This approach is relevant for digital transformation, especially if digital transformation is approached as a type of organizational change.

The framework has been developed with a pragmatic intent of making digital transformation more actionable for practitioners. When approaching the puzzle of initiating and coordinating organizational change in the pragmatic tradition, one must subscribe to a more complex and processual understanding of the world (Farjoun et al. 2015). According to the pragmatic view, practitioners and researchers alike are engaged in the process of creation and updating of meaning based on the cues they derive from the ongoing flow of experience. The knowledge that they gain from this process is practical, necessarily fallibilist, and grounded in experience (Simpson and den Hond 2022). While there is no consensus among pragmatic thinkers as to what constitutes "the truth out there," and whether it can be reached through individual or collective processes of inquiry, the general pragmatic endeavor is concerned with creating practical knowledge from previous experience (Simpson and den Hond 2022). This practical knowledge is put to use in transforming the current situation and replacing the no longer efficacious habits of thinking and acting and can be discarded when proven inadequate.

Pragmatic knowledge contributions in the field of information systems (IS) abide by the principles of referential, methodological, and functional pragmatism (Goldkuhl 2012). That means that pragmatic IS contributions must (1) create knowledge *about* action (referential pragmatism), (2) create knowledge *through* action (methodological pragmatism), and (3) create knowledge *for* action (functional pragmatism). In line with pragmatic knowledge contributions in IS, this framework was developed *about* actions and action strategies comprising digital transformation. It was developed *through* active participation in and engagement with digitally transforming organizations. Finally, the framework was developed *for* action, to help practitioners construct digital transformation in their organizations.

With digital transformation being a continuous, complex, and organization-wide challenge that is in need of additional practice-oriented research (Hanelt et al. 2021; Satwekar et al. 2024; Vial 2019), creating it in organizations requires "drastically alter[ing] the boundaries of existing knowledge, established disciplines, and available capabilities" (Winter and Butler 2011, p. 100). Approaching such a challenge relies on "collaboration among researchers across projects that may occur over many years (perhaps decades)" to "bring [in] different perspectives […] develop new practices, processes and norms and build on each other's successes" (Paradice et al. 2019, p. 323). In other words, challenges that require altering the boundaries of existing knowledge, disciplines, and capabilities are better understood through research programs rather than through a singular research project. The process of framework development, therefore, similar to the research process of Paradice et al. (2019), is characterized as a research program spanning across a multitude of projects that the primary author has been intimately involved in through (1) direct participation in the research project, and (2) direct participation through mentorship and supervision of research projects. Due to the variety and multitude of empirical inputs that went into the process of framework development, an in-depth overview of each project is beyond the scope of this paper. Therefore, this section proceeds with presenting the number and types of empirical inputs as well as the type of data that was collected (Sect. 2.1), discussing the way the data was analyzed (Sect. 2.2), and illustrating some of the consequential pivots that took place when developing each framework component (Sect. 2.3).

2.1 Data collection

This pragmatic framework has been developed between 2014 and 2024. During that period, the first author's simultaneous involvement in in-company interventions and theorizing and teaching on digital transformation allowed for the continuous generation and contrasting of insights. Additionally, being involved in research projects through direct participation and supervision added to the body of materials collected for tackling the challenge of digital transformation. While the theoretical basis of the framework is presented in Sect. 3, empirical inputs serving the development of the framework include:

(1) 18 editions of the "Leading Digital Transformation" program (once a year, 2014–2015; twice a year, 2016–2024), which was launched into the Open Executive Education Portfolio of Vlerick Business School in the Summer of 2014. Each edition of the program involved five days of full-time in-person engagement, sensemaking, and problem-solving with individuals responsible for digital transformation in their organizations. This program provided the primary author with access to and close familiarity with digital transformation challenges experienced by over 500 practitioners, the way that they were making sense of these challenges, as well as abundant testing of each of the framework components. In each run of the program, at least one person from the research team was present to take note of new and surprising observations that went into the abductive process of framework development.

(2) Written accounts of digital transformation challenges of 1000 practitioners, in the form of homework for the three online editions (yearly, 2019–2021) of a 12-week

self-paced course, named "Take the Lead in Digital Transformation." Similar to the "Leading Digital Transformation" program, the self-paced course centered around using the framework to analyze and lead digital transformation in practitioners' organizations. The homework included collecting and reporting digital transformation observations with the help of the framework. It served both informing the framework as well as testing to what extent it allows practitioners to step into the role of pragmatic researchers and designers in their organizations. Collected records from the course, including the homework assignments, further served as a basis for designing follow-on research projects and publications (e.g., Vanderheyden et al. 2025).

(3) Over 40 in-company interventions, catering predominantly to medium- and large-sized public and private organizations across diverse sectors, which were initiated because of and for the purpose of using the framework to initiate, coordinate, and maintain organizations' digital transformations. Several of the interventions culminated in teaching case studies, the materials for which were gathered through a series of structured and unstructured interviews, written-up, supplemented by teaching notes and published (e.g., Verweire et al. 2017a, 2017b; Viaene 2018, 2019, 2024) .

(4) Six organizational engagements led to privately funded PhD research conducted at KU Leuven , under the supervision of the primary author, affording opportunities for further in-depth and at-length exploration of various facets of the framework.

2.2 Data analysis

The abovementioned empirical inputs involved varied methods of data collection and analysis. Primary data gathered from all research projects together included structured and unstructured interviews, in-company observations and interventions, written and oral accounts of digital transformation experiences, and notes from class discussions. Secondary data included aggregated insights from research projects and resulting publications, shaping the researchers' progressive understanding of the digital transformation phenomenon and its challenges. The data and publications resulting from all research projects formed a pool of materials that was added to over time and that, for the purposes of framework development, was iteratively analyzed and consulted in an abductive manner.

The abductive analysis of the research materials involved going through all gathered materials iteratively. While both inductive and abductive analyses involve iteration in analyzing gathered materials, induction aims to make sense of the gathered materials through aggregation, while abduction "urges us to start sense-making from surprising observations" (van Hulst and Visser 2025, p. 569). This approach, emphasizing the breakdown of former expectations, is a process in which "researchers ask themselves, over and over, what the phenomenon they study is a case of" (van Hulst and Visser 2025, p. 569). With the phenomenon of digital transformation being relatively novel and certainly complex, with its complexity still not being given its due (Satwekar et al. 2024), abductive analysis underlay the continuous reconceptualization of digital transformation which was necessary to create actionable knowledge for organizations.

More precisely, the process of abductive theorizing takes place by researchers working in iterative loops of theorizing from data to conceptualization and vice versa "as the researcher's understanding constantly changes – ideally progressing toward a coherent account, but possibly breaking down at various moments" while "the meaning of the whole arises from the interpretation of its parts and vice versa" (van Hulst and Visser 2025, p. 571). In developing the framework, this iterative process of abductive theorizing was conducted based on the continuously growing pool of materials (e.g., articles, teaching cases, interviews, course assignments, notes from the classroom) and all the while facing an evolving digital transformation problem space.

The iterations in the process of analysis and design did not follow a structured pattern. Instead, discovering a new and revealing theoretical lens meant both going back to reinterpreting materials from the pool and conceptualizing the framework differently. Similarly, encountering a surprising observation not fitting the pre-existing framework conceptualization and its theoretical backing led to reinterpreting the framework and its theoretical background. That means that the process of the framework development was complex and not easily representable. Complexity in the research process, however, is necessary to do justice to organizational complexity (Beer 1994; Tsoukas 2017) that characterizes digital transformation (Satwekar et al. 2024). To make sure that the resulting framework not only captured organizational complexity inherent to digital transformation but also remained pragmatic, the framework underwent continuous validation. Beyond being applied and tested in each organizational intervention by the research team, the framework was evaluated at a fixed cadence during and after each iteration of the "Leading Digital Transformation" Executive Education program over the entire period of study (2014–2024), based on whether it made sense and was helpful and actionable for practitioners.

2.3 Components overview

The pragmatic framework of digital transformation leadership, presented in Fig. 1, consists of five components (A-E). A unique account of digital transformation leadership (Sect. 3) is summarized in analytical pragmatic propositions (*Component* A). This account grounds a 2×2 canvas made of axes of paradoxical organizational tensions - exploration vs. exploitation, and belief- vs. action-driven sensemaking (*Component* B; Sect. 4.1). These tensions need to be carefully managed by ensuring that the digital transformation focus on one or the other conflicting imperative is an organizations' conscious choice, and all conflicting aspects are given due attention. To manage these paradoxical tensions, the canvas proposes a pragmatic researcher or a reflective practitioner to analytically distinguish four digital transformation practice areas – "vigilant," "voyager," "visionary," and "vested" – that need to be cultivated and balanced in organizations. These practice areas represent four contexts in which digital transformation efforts need to unfold. To further make the practice areas of digital transformation more actionable, the framework includes digital transformation progress points for each practice area ("jobs to be done") (*Component* C; Sect. 4.2). To assist in translating these progress points into digitally-driven ways of working, the framework synthesizes "digital transformation leadership qualities" pointing towards behavioral qualities that consistently contribute to the success of new prac-

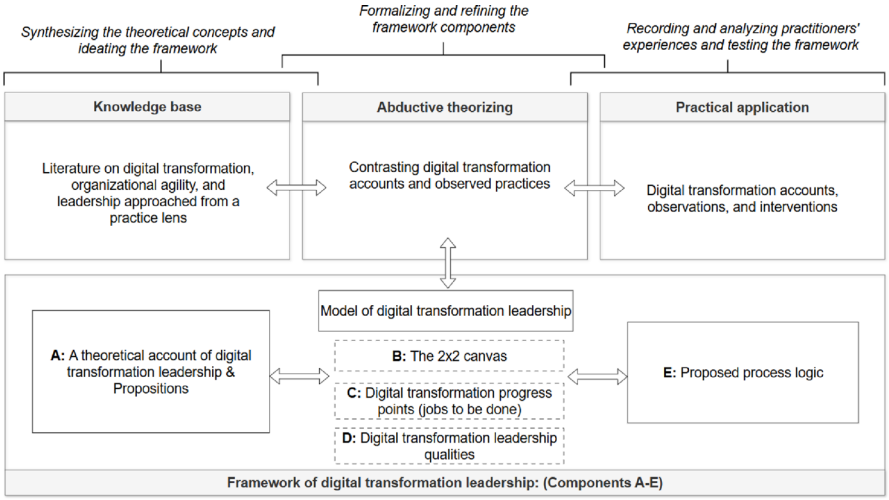

Fig. 1 Framework development through abductive theorizing

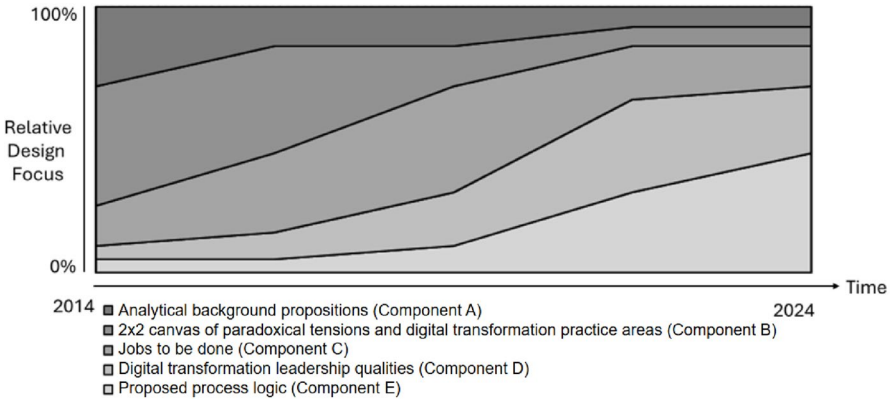

Fig. 2 Shifting design focus across framework components over time (2014–2024)

tices in each digital transformation practice area (*Component* D; Sect. 4.3). The 2 × 2 canvas of digital transformation paradoxical tensions and practice areas (*Component* B; Sect. 4.1), jobs to be done or the lower-level digital transformation progress points (*Component* C; Sect. 4.2), and digital transformation leadership qualities capturing the complementary motivational and behavioral components of practices (*Component* D; Sect. 4.3) constitute what we call *the model of digital transformation leadership* (Sect. 4) that sits at the heart of the framework. Finally, the framework includes a process logic for operationalizing it in organizations (*Component* E; Sect. 5).

As presented in Fig. 2, the focus of designing and readjusting the framework has been shifting across the five framework components throughout the different editions of the teaching programs and company interventions over the years. The theoretical account of digital transformation leadership and background propositions (*Compo-*

nent A; Sect. 3) emerged while bringing together (1) discussions with practitioners and analysis of their digital transformation experiences, (2) research on digital transformation, and (3) literature on organizational change. Practitioners' accounts and research reflected the conceptualization of digital-age organizational agility as the primary goal of digitally transforming organizations. Analyzing practitioner accounts revealed the practical struggle of initiating and maintaining the collective action necessary for digital transformation. Practitioners in question lacked understanding and tools for initiating, maintaining, and governing the emergence and establishment of new digital ways of working constituting digital transformation. In the theoretically derived account and propositions (*Component* A; Sect. 3), we made sense of and conceptualized these missing organizing processes as digital transformation leadership. Therefore, the theoretical account and propositions reflected researchers' sensemaking of digital transformation over the years that grounded the rest of the framework, with the current formulation established in 2024.

The 2 × 2 canvas (*Component* B; Sect. 4.1) defining practice areas of digital transformation emerged early on, following from the theoretical account of digital transformation and practitioners' experiences. While some of the changes made to the canvas turned out to be inconsequential, others translated into the final state of the framework. For example, an earlier interpretation of the axes in the framework (e.g., Viaene 2017, 2018, 2020) was significantly refined. That happened as the pool of materials informing the framework expanded, drawing, among others, on knowledge from a research collaboration with a global design and engineering company on addressing organizational tensions in digital transformation (Danneels & Viaene 2022). Reinterpreting the axes of the canvas as axes of organizational tensions proved more revealing in making sense of practitioners' challenges.

With digital transformation understood as continuous organizational change, the canvas then displays two paradoxical tensions that are reflected in the literature and captured in practitioners' accounts, that need to be consciously balanced by organizations. In light of resource constraints, organizations must balance their focus on transforming their capacity for exploration and their capacity for exploitation. At the same time, organizations need to similarly balance pushing what is thinkable and what is actionable for them in terms of new digital opportunities. No matter the balancing choices made, however, organizations need to make sure that both capacities and both modes of approaching new digital opportunities contribute to their digital transformation cohesively. Identifying these exploration vs. exploitation and belief- vs. action-driven sensemaking paradoxical tensions (Sects. 3, 4, and 5) helped organizations to analyze their digital transformation experiences. That was done to reflect whether their balance of exploration- vs. exploitation-oriented activities as well as belief- or action-driven sensemaking was indeed consciously chosen.

While the 2 × 2 canvas (*Component* B; Sect. 4.1) was used to analyze digital transformation initiatives and new digitally-driven ways of working in terms of dealing with the tension axes, lower-level digital transformation progress points (*Component* C; Sect. 4.2) started to emerge. While it is impossible to comment on all changes that led to the emergence of this framework component, the early stages of applying the framework revealed that making sense of digital transformation at the level of the four contexts in which digital transformation efforts take place was not detailed enough to

guide practitioners. At that stage, without yet having a clear concept to work with but needing to distinguish "what digitally transforming organizations were aiming for at a more granular level," the research team intensified its efforts to identify, document, and understand the ways of working introduced as well as their rationale under the banner of digital transformation. Newly observed changes, together with those identified from the pool of existing materials, were inductively generalized into progress points – descriptive themes signaling progress or success – situated within the four contexts of the canvas. The notion of a job to be done was introduced later, allowing this framework component to resonate more strongly with practitioners as well as with the intent of the research (*Component* C; Sect. 4.2).

As a result, the goals of all discussed digital transformation initiatives and newly introduced ways of working were combined and conceptualized into jobs to be done (*Component* C; Sect. 4.2), that is, aims or progress that organizations need to strive to achieve when introducing new digitally-driven ways of working. The experiences of digitally transforming organizations were synthesized into goals rather than presented as ways of working due to suitable ways of working in achieving a job to be done being highly varied and context-dependent. While orienting themselves towards lower-level digital transformation progress points, organizations still need to develop and establish particular ways of working suitable for their organization, making their digital transformation appropriate for them, and thus more sustainable over time. This account illustrates how meaning and practical utility emerged and were iteratively refined for this framework component until they received repeated validation from practitioners.

While analyzing the establishment of new digitally agile ways of working, we also observed and noted down the behavioral qualities that consistently contributed to the success of new ways of working performed in each of the four practice areas. Those were the qualities that showed up most vividly and constructively in each of the practice areas successfully bringing about the change of practices. These behaviors and attitudes were gathered and described not to discriminate or reflect the reality of digital transformation but to serve as a tool to indicate the need for (developing) certain attitudes and behaviors that complement the development and performance of new practices in one or another digital transformation practice area. These qualities were not exclusive to each practice area, but they highlighted ways of human expression that most naturally supported the kind of progress each practice area called for. The descriptions of supporting behaviors and attitudes for each digital transformation practice area found resonance with practitioners.

While the extracted attitudes and behaviors and the research process behind them underwent many changes, establishing a correspondence between changes to ways of working and practices led to reinterpreting the framework and the qualities from the lens of practice theory (Bourdieu 1990; Reckwitz 2002; Schatzki et al. 2001; Whittington 2011). This reinterpretation resulted in realizing that to create organizational conditions for continuously changing organizational practices, only defining digital transformation progress points is not sufficient. It is important for practitioners to recognize that changes in ways of working are not only goal-oriented but also involve emotional and motivational factors (Reckwitz 2002).

This realization prompted another iteration of framework development, focusing on digital transformation leadership qualities derived from available materials, informed by practice theory, and tested with practitioners during the course. As a result, observed qualities facilitating digital transformation have been similarly aggregated to make for a practical tool that can be used by individuals and organizations to guide their digital transformation efforts – digital transformation leadership qualities (*Component* D; Sect. 4.3). While the aims of practices and the complementary attitudes and behaviors have been analytically decomposed, they have been observed in practice as a joint effort driving the change in ways of working. With these qualities synthesized, it is up to individuals and organizations to check whether and how they can bring these attitudes and behaviors into developing new ways of working in digital transformation practice areas and how they can further support and encourage them if necessary.

As the interest of digitally transforming organizations shifted from initiating digital transformation towards coordinating and resolving paradoxical tensions between different digital transformation initiatives, the process logic (*Component* E; Sect. 5) went through the most active stage of its development. Using the framework in digitally transforming organizations over time revealed the need for explicit orchestration processes, even if all digital transformation practice areas are present and active in organizations. The complementary process logic evolved in parallel with the rest of the framework, drawing on the literature on organizational tension management and communities of practice and was experimented with and tested in digitally transforming organizations. The goal of the process logic was to coordinate the work across the canvas and across the organization to ensure explicit tension resolution so that digital transformation leadership can be continually developed and digital transformation can be maintained over time.

3 Background: developing a conceptualization of digital transformation leadership

In this section, in several steps, we build up the conceptualization of digital transformation leadership as a system that drives and supports digital transformation in organizations. Each step is punctuated by an analytical background proposition, translating descriptive knowledge into more actionable statements grounding the development and the application of the framework. First, to make digital transformation more actionable, we propose to view digital transformation as a process of continuous organizational change aiming toward organizational agility in the digital age. Second, we propose to view organizational change through the lens of changing practices constituting an organization, with change in practices relying on processes of organizational sensemaking. Third, we argue that continuous digital transformation depends on establishing new practices while balancing paradoxical tensions inherent to digital transformation. Finally, we propose that digital transformation requires coordinated collective action of developing and establishing new practices. This collective action, in line with the leadership-as-practice views (Carroll et al. 2008; Raelin 2016, 2023), we conceptualize as digital transformation leadership constituting

the goal and the benchmark of the framework's successful application presented in this paper.

3.1 Digital transformation as a continuous organizational change toward digital-age organizational agility

Digital transformation has been in the spotlight of strategic information systems (IS) academic and practitioner discussions in the last several years (Danneels and Viaene 2022; Hanelt et al. 2021; Omol 2024; Van Veldhoven and Vanthienen 2022; Verhoef et al. 2021; Viaene 2020; Vial 2019). Many efforts were made to analyze and provide grounding concepts for its further understanding and research (Annosi et al. 2020; Gong and Ribiere 2021; Hanelt et al. 2021; Liu et al. 2021; Omol 2024; Vial 2019). These attempts take digital transformation to be divergent from other, similarly technology-enabled, organizational transformations.

Digital-age competitive environments are characterized by the relentless progress of digital technologies changing the nature and scope of competition as well as the requirements for businesses to remain relevant (Omol 2024). The rapid and continuous progress of digital technologies makes it more important for businesses to proactively identify and respond to changes. In other words, it is increasingly important for organizations to develop and maintain their agility in the digital age (Verhoef et al. 2021; Vial 2019). Organizational agility is "a core competency, competitive advantage, and differentiator that requires strategic thinking, an innovative mindset, exploitation of change and an unrelenting need to be adaptable and proactive" (Harraf et al. 2015, p. 675). Organizational agility refers to the ability of an organization to "meet rapidly changing needs and other external stimuli cost-effectively without compromising the quality of its products and services" (AlNuaimi et al. 2022, p. 638). An agile organization in the digital age is, thus, capable of tackling challenges stemming from or catalyzed by rapidly emerging new technologies (AlNuaimi et al. 2022). In light of the progressing digital technologies disrupting market environments, agility is not a matter of choice but a matter of survival for organizations (Harraf et al. 2015). With that in mind, we propose to formulate the relationship between digital transformation and organizational agility in the way summarized by *Proposition* #1:

Proposition #1 *Digital transformation is undertaken to develop and maintain digital-age organizational agility, or an organization's ability to be a relevant actor in a digitally disruptive environment.*

Organizational agility, however, can never be completely achieved. It is an ongoing process and "more a matter of becoming than being" (Harraf et al. 2015). With agility being closely related to responding to external stimuli and rapidly changing needs, this is even more true in light of the relentless progress of digital technologies. If digital transformation is undertaken to develop and maintain organizational agility, it cannot be viewed as a one-off change. Responses to external crises or disruptions are normally associated with one-time radical changes followed by a period of relative stability (Weick and Quinn 1999). Technology-triggered organizational transformations would then be seen as episodic event changes, "punctuated by occa-

sional external events, [and] affecting internal deep structures" (Hanelt et al. 2021, p. 1177). However, the ongoing progress of digital technologies giving rise to constant changes in organizations' environments requires organizations to continually adjust to remain competitive. The periods of relative stability associated with transformation, therefore, are still characterized by smaller yet significant local changes emerging from improvisations and experimentations with technology (Hanelt et al. 2021; Verhoef et al. 2021; Vial 2019). The ongoing progress of digital technologies giving rise to changes in organizations' environments requires organizations to continually adjust to remain competitive. To keep up with responding to the constant stream of opportunities and threats created by digital technologies, organizations have to rethink their foundation in light of and with the help of digital technologies (Cennamo et al. 2020; Sebastian et al. 2017). Thus, due to the continuous nature of digital disruptions, we propose to view digital transformation as a process of continuous organizational change in light of and with the help of digital technologies, as formulated in *Proposition* #2:

Proposition #2 *Digital transformation in pursuit of digital-age organizational agility is a process of continuous organizational change in light of and with the help of the ongoing advancement of digital technologies.*

Connecting *Propositions* #1 & #2, and emphasizing that the success of digital transformation depends on an organization's ability to remain agile in the face of ongoing digital disruption, we propose to define digital transformation as *a process of organizational change constituting a strategic response to the environmental changes invoked by a continuous stream of digital opportunities and challenges.*

3.2 Digital transformation as a change of practices grounded in organizational sensemaking

Approaching digital transformation as continuous organizational change and developing (pragmatic) knowledge on it, requires one to specify their understanding of organizations and change. Approaching the subject of organizational change, thus, requires one to give up a largely static and fixed view on organizations (Farjoun et al. 2015). One way to bring change and process into the picture, in line with the pragmatic view, is to take organizations to be in a constant flux, or an ongoing state of becoming (Farjoun et al. 2015). In this line of thinking, organizational structures, shaping the course of organizational activity, are themselves constituted by events that are more regular and rhythmic compared to other events that are more irregular and spontaneous (Farjoun et al. 2015). Understanding an organization in terms of events performed over time is in line with the "practice turn" in organizational theory, focusing on practices as a unit of analysis in organizations (Simpson 2009). Taking a practice-based and processual perspective on organizations allows explaining both organizations changing and persisting over time.

Taking center stage in practice theories (Schatzki et al. 2001), a practice is seen as "a routinized way in which bodies are moved, objects are handled, subjects are treated, things are described and the world is understood" (Reckwitz 2002, p. 250).

It is a "routinized type of behaviour which consists of several elements, interconnected to one other: forms of bodily activities, forms of mental activities, 'things' and their use, a background knowledge in the form of understanding, know-how, states of emotion and motivational knowledge" (Reckwitz 2002, p. 249). Described as a "nexus of doings and sayings," a practice, comprising a way of doing something and consisting of specific interconnected elements, is understandable not only to the performer of it but also to the potential observers situated in the same cultural context (Reckwitz 2002, p. 250). Taking practices to be constituents of organizations and of the social world at large, "the social [becomes] a field of embodied, materially interwoven practices centrally organized around shared practical understandings" (Schatzki et al. 2001, p. 12). It is still up to debate whether, even within the practice view, there are elements in organizational ontology beyond practices. However, just focusing on practices, connecting material factors with understanding, including know-how but also states of emotion and motivational knowledge, already allows a pragmatic researcher and a reflective practitioner to better understand how organizations change and create pragmatic knowledge on how this change can be facilitated, which is important for the digital transformation domain.

Approached from a processual and practice-based view, digital transformation can be reinterpreted from a vague process of organizational change into an ongoing process of introducing new and changing practices in an organization to make the organization more aligned with its digitally disruptive environment. At the same time, establishing these practices relies on actively and continually figuring out new relationships between people and technology as technological advancements continue to afford increasing action- and decision-making autonomy on the side of algorithms and machines (Brynjolfsson 2022; Cremer and Kasparov 2021; Duan et al. 2019). In line with this understanding, we propose that the continuous introduction of new "digitally agile" practices or digitally agile ways of working into the fabric of the organization, then, constitutes digital transformation, as summarized in *Proposition* #3:

Proposition #3: *Digital transformation is constituted by continuously changing established practices or ways of working that require discovering and figuring out new relationships between people and technology.*

Understanding digital transformation at the level of a change of practices makes it more approachable in practice and for pragmatic research. This understanding allows for thinking about and engaging with change on the level of (1) individuals and groups, and (2) through engaging with components of practices, including material factors as well as shared understandings grounding and interconnected with them (Schatzki et al. 2001). The objective of changing practices and, therefore, processes of organizing, requires further elaboration on how processes of organizing, in our case taken to be presented by practices, unfold.

In the influential account of Weick (1995), organizing is a way to reduce the complexity of the changing world. The world abounding with the possibility for action under the conditions of limited knowledge is equivocal – that is, situations can be interpreted in many plausible ways which can delay or disrupt coordinated action.

This equivocality is resolved in socially created (enacted) environments through "sensible interlocked behaviors" (Weick 2020). That happens as people, confronted with equivocality, comprehend a situation explicitly in words within their social context, with this created understanding serving as a "springboard for [coordinated and orchestrated organizational] action." Establishing shared understandings through processes of collective organizational sensemaking, therefore, contributes to and grounds reorganizing through practice change.

Organizational sensemaking, more specifically, is a process of the social construction of meaning in light of discrepant cues that interrupt ongoing activity (Maitlis and Christianson 2014; Weick 1995). It is characterized by creating shared understandings of disruptions to an extent that "enable[s] the same behavioral consequences" (Maitlis and Christianson 2014, p. 561) and allows for restoring organizational activity. With digital transformation requiring new behavioral consequences, or new practices to be continuously developed and established, digital transformation would involve continuously stimulating and engaging people in the process of organizational sensemaking. This understanding is summarized and captured in *Proposition* #4:

Proposition #4: *Digital transformation requires continuous sensemaking of new digital opportunities to serve as a springboard for new practices resulting in reorganizing for more digital-age agility.*

3.3 Digital transformation as continuous management of paradoxical tensions

Summarizing: Digital transformation taken to be a persistent development of new ways of working for digital-age agility (*Propositions* #1 & 3) requires a continuous shift of practices. Maintaining this shift of practices requires continuous creation of new shared meanings allowing to reorganize for more digital-age agility (*Proposition* #4). Achieving continuity in digital transformation, necessary for digital-age agility, however, depends on balancing paradoxical organizational tensions inherent to digital transformation (Danneels and Viaene 2022; Soh et al. 2023).

As established in Sect. 3.1, organizational agility refers to the ability of an organization to "meet rapidly changing needs and other external stimuli cost-effectively without compromising the quality of its products and services" (AlNuaimi et al. 2022, p. 638). This description reveals the balance that needs to be maintained in light of the implied resource constraints. Remaining agile as an organization in the digital age requires sensing environmental changes and acting on them on a permanent basis. The ability to take timely action, however, is costly, as it involves having scarce resources on standby to prepare for potential environmental changes (Teece et al. 2016). This balance has already been discussed in the context of the related concept of ambidexterity (Armstrong and Manitsky 2022), attributed to (March 1991).

Ambidexterity refers to the balance that organizations need to strike in allocating their scarce resources between exploitation and exploration activities (March 1991). March (1991, p. 71) characterized organizations' exploitation activities as those focusing on "refinement, choice, production, efficiency, selection, implementation, [and] execution." Exploration activities, in their turn, would include "search, variation, risk-taking, experimentation, play, flexibility, discovery, [and] innovation" (March

1991, p. 71). To maintain continuity in digital transformation, organizations need to change digital practices (relying on developing new shared meanings) for injecting agility into both their exploration and exploitation capacities. Limited resources as well as the need to maintain the quality of products and services will likely translate into prioritizing digitally transforming one capacity over another. Over time, however, to develop digital-age organizational agility, both capacities need to be up to date to effectively operate in the changing competitive environment. Digital transformation, therefore, requires organizations to manage the tension between transforming their exploration and exploitation capacities (Andriopoulos and Lewis 2009; Lin et al. 2013). This tension is paradoxical, as it cannot be resolved by choosing one of the contradictory elements to define overall success. A digitally transforming organization cannot make a choice between transforming its exploration or exploitation capacity only. The paradoxical tension, then, is constituted by exploration and exploitation comprising "contradictory yet interrelated elements that exist simultaneously and persist over time" (Smith and Lewis 2011, p. 386).

The necessity to manage the paradoxical exploration-exploitation tension in digital transformation is summarized in *Proposition* #5:

Proposition #5: *Achieving digital-age organizational agility requires organizations to manage the paradoxical tension between digitally transforming their exploration and exploitation capacities in light of resource constraints.*

As established in Sect. 3.2, developing new practices constituting digital transformation requires maintaining continuous organizational sensemaking of new digital opportunities. What is also the case, is that organizational sensemaking can take place in predominantly belief- or action-driven ways (Weick 1995). When faced with a disruptive event, individuals can approach it by first building an interpretation of it based on their prior beliefs, and then acting on the basis of their interpretation. Alternatively, individuals can act first, thereby gathering more information about the event, and building its interpretation retrospectively (Choo 2002). Approaching disruptions in predominantly belief- or action-driven ways comes with another trade-off. Belief-driven sensemaking allows for more deliberate and considered actions, although the interpretation grounding them might not be accurate anymore and the action itself is delayed. Action-driven sensemaking is more risky when the "appropriate" course of action is not yet clear. However, by acting first, one can shape the course of events as they unfold and gather more accurate information for further responding to the disruption (Weick 1995).

While belief- and action-driven sensemaking naturally feed into one another, to stimulate organizational sensemaking, organizations can engage in activities explicitly contributing to one or the other type of sensemaking. Activities can predominantly contribute to organizational sensemaking via belief if they focus more on creating shared meaning through exchanging ideas and opinions (van der Heijden et al. 2010), or can be driven predominantly via action, while "developing collective activities in order to generate a shared meaning" (van der Heijden et al. 2010, p. 1790). Encouraging both types of sensemaking can ensure that the organization as a whole does not approach disruptions in a way biased towards belief- or action-

driven sensemaking. For example, in organizations' decision-making contexts, a bias towards belief-driven sensemaking can lead to "paralysis by analysis" and a bias towards action-driven sensemaking can lead to "extinction by instinct," with a fine line needing to be threaded between the two (Langley 1995).

The tension between activities stimulating belief- or action-driven sensemaking, in the context of organizations' digital transformations cannot be resolved in favor of one of the contradicting imperatives. That is due to the fact that organizations need to continuously learn about and experiment with new technologies while also figuring out how they can be embedded into ways of working that are in line with the existing shared meanings within the organization. Therefore, continuously developing new ways of working to reorganize for increased digital-age organizational agility requires organizations to stimulate both belief- and action-driven sensemaking of new digital opportunities, meaning that the tension between belief- and action-driven sensemaking is paradoxical. The necessity to manage the paradoxical tension between stimulating belief- and action-driven sensemaking in digital transformation is captured by *Proposition* #6:

Proposition #6: *Achieving digital-age organizational agility requires organizations to manage the paradoxical tension between activities stimulating belief- and action-driven sensemaking of new digital opportunities in light of biases that organizations can display towards one or the other.*

One way for organizations to manage paradoxical tensions is by first separating the conflicting imperatives to give each their due recognition, and then forging connections between them "to leverage interdependencies and benefit from their synergies" (Smith et al. 2016). Managing explore-exploit and belief-action paradoxical tensions in a digitally transforming organization would then involve identifying the goals behind each of the conflicting elements and balancing the focus on each of them. At the same time, since both opposing elements are necessary and can reinforce each other in digitally transforming the organization, connections need to be established between the focus areas to allow for interdependencies and virtuous cycles to emerge (Smith and Lewis 2011). As summarized in *Proposition* #7, maintaining the continuity of digital transformation, therefore, involves managing paradoxical tensions which require developing creative ways to engage both contradictory elements without sacrificing or undermining the goals of either.

Proposition #7: *Continuous digital transformation requires managing paradoxical organizational tensions through developing creative ways to engage and connect contradictory focus areas to allow for synergies and virtuous cycles to emerge.*

3.4 Digital transformation as driven by digital transformation leadership

From the accounts put forward in the previous subsections, digital transformation emerges as a strategic and continuous effort (*Proposition* #2) to stimulate organizational sensemaking of digital disruptions (*Proposition* #4) to allow for developing new ways of working (*Proposition* #3) that leverage digital technologies for explora-

tion and exploitation activities (*Proposition* #5) to make an organization more digital-age agile (*Proposition* #1). In the process, organizations need to develop creative ways and systems to engage and connect different focus areas (*Propositions* #5, 6, & 7). This effort, involving systematically creating, connecting, and managing new ways of working, cannot be approached from a purely individualistic or causal competency-based view of leadership or a top-down approach to introducing changes.

Establishing new agile practices involves routinizing agile "ways of understanding, knowing how, ways of wanting and of feeling that are linked to each other within a practice" (Reckwitz 2002, p. 253). It is rooted in creating a shared understanding within a given cultural environment. Furthermore, it requires developing "a particular routinized mode of intentionality, i.e. of wanting or desiring certain things and avoiding others" (Reckwitz 2002, p. 254). Establishing new practices in uncertain environments involves going beyond promoting particular methodologies or behaviors. It requires individuals to collaboratively make sense of – craft a shared understanding of – new practices and their purpose as a community. This endeavor requires each individual to find ways to participate in a "process of intelligent adaptation, renewal, and legitimate change" (Farjoun et al. 2015, p. 1794) by sensing new opportunities, breakdowns of old structures, and using them as "leverage and scaffolds for improvisation, bricolage, and transformation" (Farjoun et al. 2015, p. 1794).

Putting in place systems for continuously recognizing new digital opportunities and introducing new and changing ways of working requires this mode of intentionality to be spread across the organization. The collective enactment of these systems requiring a new "transformational" mode of intentionality and resulting in transforming the organization over time is what we propose to call digital transformation leadership. *Proposition* #8, below, summarizes the conceptualization of digital transformation leadership proposed in this paper:

Proposition #8: *Digital transformation leadership is constituted by the enactment of systems of continuously establishing new practices organization-wide, requiring a new mode of intentionality, and resulting in reorganizing for more digital-age agility.*

This view of leadership as collaborative agency stretched over its participants is aligned with the leadership-as-practice theories, taking leadership to be situated in an organizational context and collectively and relationally constructed in day-to-day activities and interactions (Carroll et al. 2008; Farjoun et al. 2015; Raelin 2016, 2023). Leadership-as-practice scholarship investigates manifestations of leadership in organizational practices with leadership itself being the subject of investigation (Raelin 2020). We define digital transformation leadership in *Proposition* #8, as a similarly emergent property, but one that is desirable to set into motion and drive continuous and context-dependent digital transformation. This digital transformation leadership, requiring a shared mode of intentionality aimed at making the organization more digitally agile over time, is the goal and the benchmark of the successful application of the digital transformation leadership framework presented in this paper. The purpose of conceptualizing digital transformation leadership is to make current research and theorizing on leadership-as-practice applicable to the digital transformation domain. The correspondence between the claim made in leadership-

as-practice theorizing and the analytical propositions making it actionable for digital transformation, is presented in Table 1. The table provides a similar correspondence between discussed descriptive claims and analytical background propositions for each of the eight formulated propositions.

Table 1 Pragmatic propositions presented against the descriptive claims grounding them

Descriptive claims:	Analytical background propositions:
Digital transformation is undertaken to remain aligned with an organization's digitally disruptive competitive environment. The ability to remain aligned with the digitally disruptive competitive environment can be understood as organizational agility in the digital age.	***Proposition*#1**: Digital transformation is undertaken to develop and maintain digital-age organizational agility, or an organization's ability to be a relevant actor in a digitally disruptive environment.
Organizational agility is a process rather than a static state.	***Proposition #2***: Digital transformation in pursuit of digital-age organizational agility is a process of continuous organizational change in light of and with the help of the ongoing advancement of digital technologies.
Organizations are composed of socially constructed and enacted practices. Digital technologies increasingly shift the balance of agency between humans and machines.	***Proposition*#3**: Digital transformation is constituted by continuously changing established practices or ways of working that require discovering and figuring out new relationships between people and technology.
The emergence of new practices in organizations requires new meanings and new shared understandings.	***Proposition*#4**: Digital transformation requires continuous sensemaking of new digital opportunities to serve as a springboard for new practices resulting in reorganizing for more digital-age agility.
Organizational agility refers to the ability of an organization to meet new demands without compromising on its current operations. Organizational activities can be analytically split into those that are exploitation- or exploration-oriented. In light of the resource constraint, organizations need to balance the focus on developing their exploration or exploitation capacities.	***Proposition*#5**: Achieving digital-age organizational agility requires organizations to manage the paradoxical tension between digitally transforming their exploration and exploitation capacities in light of resource constraints.
Organizational sensemaking can take place in predominantly belief- or action-driven ways. Organizations can display a bias towards approaching disruptions primarily through stimulating one or the other type of sensemaking.	***Proposition*#6**: Achieving digital-age organizational agility requires organizations to manage the paradoxical tension between activities stimulating belief- and action-driven sensemaking of new digital opportunities in light of the bias that organizations can display towards one or the other.
Paradoxical tensions are most effectively managed by looking for and tapping into synergies between the opposing elements.	***Proposition*#7**: Continuous digital transformation requires managing paradoxical organizational tensions through developing creative ways to engage and connect contradictory focus areas to allow for synergies and virtuous cycles to emerge.
Leadership is an emergent property of social systems that manifests itself in social and constructed practices shared by its participants.	***Proposition*#8**: Digital transformation leadership is constituted by the enactment of systems of continuously establishing new practices organization-wide, requiring a new mode of intentionality, and resulting in reorganizing for more digital-age agility.

4 Modelling digital transformation leadership

Based on the synthesized account presented in the previous section, digital transformation emerges as a continuous journey of introducing and changing practices through developing new shared practical understandings for the purpose of digital-age organizational agility (*Propositions* #1–6). This continuous journey relies on individuals enacting systems of continuous establishment of new practices while creatively connecting areas of potential paradoxical tensions (*Propositions* #7 & 8). This enactment of change required for digital transformation is the work of digital transformation leadership, requiring voluntary participation on the side of organization members imbued with a new "transformational" mode of intentionality (*Proposition* #8). The presented account of digital transformation and digital transformation leadership has been formulated to ground the model of digital transformation leadership which comprises a collection of practical tools for developing digital transformation leadership in organizations. This model of digital transformation leadership lies at the core of the framework presented in this paper.

The model of digital transformation leadership is meant to be used to (1) help organizations identify digital transformation practice areas giving rise to tensions, (2) provide guidance to the process of practice development in terms of its purpose (how it adds to organizational agility) and its scope (how new practical understandings are stimulated), as well as (3) encourage the development of practices by illustrating complementary behavioral qualities that contribute to their success.

This section presents the components of the digital transformation leadership model as used in practice and comprising two levels of conceptualization: (1) a 2 × 2 canvas of digital transformation paradoxical tensions and practice areas (*Component* B), and (2) lower-level digital transformation progress points ("jobs to be done") (*Component* C) and complementary behavioral qualities that contribute to their success ("digital transformation leadership qualities") (*Component* D). All parts of the model constitute practical tools that, with sufficient socialization, can be and are used by practitioners to facilitate the emergence of digital transformation leadership for driving digital transformation. A proposed process logic for operationalizing the model in organizations (*Component* E) is discussed in Sect. 5, thus, completing the presentation of all framework components.

4.1 2 × 2 canvas of digital transformation paradoxical tensions and practice areas

The highest layer of model conceptualization is constituted by a theoretically grounded 2 × 2 canvas of digital transformation paradoxical tensions and practice areas. The horizontal "explore-exploit" axis captures the paradoxical tension arising in transforming the explorative and exploitative capacities of organizational agility (*Proposition* #5). The vertical "belief-action" axis reflects the paradoxical tension between approaching the transformation by engaging in predominantly belief- or predominantly action-driven organizational sensemaking (*Proposition* #6). The 2 × 2 canvas, therefore, analytically separates activities of digitally transforming organizations into four categories based on their focus on exploration or exploitation and on prompting organizational sensemaking of digital transformation in a belief- or

action-driven way. Agile practices can accordingly be categorized in relation to the horizontal and the vertical axes, reflecting, respectively, (1) their "transformational purpose" and (2) their "transformational scope." Transformational purpose qualifies whether progress with the introduction of a new way of working is sought by introducing digitally agile practices in the space of the organization's exploration or exploitation capacity. Transformational scope qualifies whether new digitally-driven ways of working are primarily introduced via belief- or action-driven sensemaking. For the purpose of making the meaning more transparent for practitioners, belief-driven sensemaking has also been labeled as "pushing the edge of the thinkable," as action-driven sensemaking as "pushing the edge of the actionable."

The canvas, presented in Fig. 3, is meant to be used for qualifying and categorizing the establishment of agile practices (*Proposition* #3). The agile practices that are being developed, thus, can be interpreted in terms of their focus on delivering digital transformation progress in one of the four practice areas outlined by the canvas, which makes it easier to keep track of and manage digital transformation. Applied to organizations' digital transformation experiences, the canvas, thus, can be seen to define four areas of agile practices and digital transformation leadership, which are named "vigilant," "voyager," "visionary," and "vested." These areas are presented for practical use in the following way.

Vigilant practices transform an organization's exploration-focused activities in light of and with the help of digital technologies. They stimulate organizational sensemaking of new digital opportunities with a drive for reconnecting the thinking dots and pushing the edge of what is thinkable for the organization. Voyager practices also purposefully employ digital technologies to transform and facilitate the organization's exploration of new digital opportunities. Contrary to vigilant practices, however, their transformational sensemaking scope is focused on reconnecting the action dots and pushing the edge of what is actionable for the organization. Visionary practices, like vigilant practices, aim to shift minds in the organization. Unlike vigilant practices, visionary practices focus on reshaping the organization's exploitation logic; that is, redefining how the organization scales digital opportunities for purposeful value creation and organizational performance. Like visionary practices,

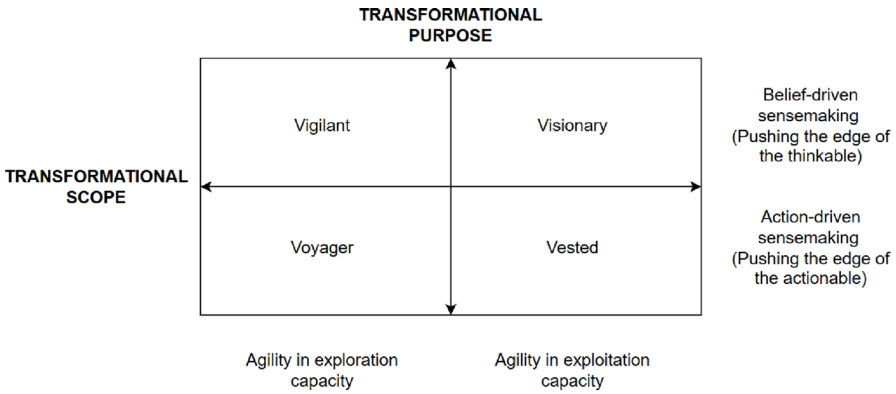

Fig. 3 2 × 2 canvas of digital transformation paradoxical tensions and practice areas

the purpose of introducing vested practices is to cater to and enhance organizations' exploitation activities. Unlike visionary practices, however, their scope is set to reconnect the action dots for the organization and push the edge of the actionable.

4.2 Lower-level digital transformation progress points ("jobs to be done")

The second layer of the model conceptualization (*Components* C & D), framed by the 2 × 2 canvas of digital transformation paradoxical tensions and practice areas, is constituted by our analysis of observed digital transformation practices and the way they are developed in organizations. Laying the 2 × 2 canvas over the experiences of digitally transforming organizations and analyzing their activities based on their vigilant, voyager, visionary, or vested focus resulted in identifying practices introduced for the purpose of digital transformation. These practices have been analyzed in order to identify and generalize their aspired progress points in each digital transformation practice area ("jobs to be done") (*Component* C) but also to capture their complementary motivational and behavioral aspects ("digital transformation leadership qualities") (*Component* D; Sect. 4.3).

When analyzing digital transformation practices with respect to their aspired progress, the 2 × 2 canvas (*Component* B; Sect. 4.1) served to generalize the recorded practices attributed to each practice area. This generalization was based on identifying and meaningfully aggregating the actual progress that practices aimed to achieve, for which we employed the concept of "jobs to be done," which was introduced and popularized by Christensen and Raynor (2003). The authors take a job to be done to be a better way for companies to understand customer needs, with jobs to be done representing the "why" of customers or the need that they want to fulfill with their purchase of a good or service. The idea is that rather than buying that good or service customers are willing to pay for progress in addressing their needs that this good or service represents.

In the context of organizations' digital transformation, we take a job to be done to be the agile progress that organizations are seeking, with "customers" being organizational stakeholders, the "product" being the new state of an organization, and digital transformation practices working toward delivering that product. For example, one of the organizations informing this research engaged in product and process portfolio simplification as part of their digital transformation journey. In this case, it helped to rethink the organization's core organizational capabilities and operating model for achieving digital-age customer-centricity, which was identified as one of the overarching jobs to be done in the model (job to be done V3.2 in Table 2). In other words, introducing practices is a means to an end; an end that is specified and generalized using the concept of a job to be done, and contextualized and categorized at the highest level by the description of the quadrant in the model, as per the model's axes. Employing the jobs to be done concept allowed for developing a deeper understanding of the transformational purpose that practitioners seek in different digital transformation practice areas.

Table 2 summarizes the generalized jobs to be done identified for each practice area of the model, reflecting the agile progress points to achieve in each digital transformation practice area. Presented this way, the list of jobs to be done captures prog-

Table 2 Jobs to be done of digital transformation leadership types

Leadership Type	Jobs To Be Done Introduction of agile practices in each area aims at:
Vigilant (V1)	V1.1 Keeping people alert for new digital-age opportunities and threats beyond the boundaries of the organization or sector. V1.2 Helping people make sense of what new digital opportunities and threats could mean for them. V1.3 Inspiring people to explore new ways of working powered by digital technologies.
Voyager (V2)	V2.1 Bundling together the diversity and creativity of individuals into an entrepreneurial team. V2.2 Exploring solutions by efficiently progressing through a steadily paced process of build-measure-learn feedback cycles. V2.3 Using empirical feedback from customers and other stakeholders to drive forward the exploration of solutions.
Visionary (V3)	V3.1 Telling a purposeful and inspirational transformative story of competitive advantage advocating the empowering use of digital technologies. V3.2 Boldly rethinking the organization's core capabilities and operating model for achieving digital-age customer-centricity. V3.3 Envisioning success by combining the organization's critical digital assets and those of ecosystem partners.
Vested (V4)	V4.1 Keeping the entire organization, rather than individual elements, on a roadmap to successful digital transformation. V4.2 Mobilizing the right skills and resources for timely exploitation and exploration. V4.3 Facilitating the learning and adoption of digital technologies, work and organizing practices at scale for an empowered workforce.

ress points defined for the four practice areas that need to be addressed by digitally transforming organizations. By taking on a job to be done in one or another area of the canvas, therefore, practitioners can effectively contribute to leadership driving their organization's digital transformation (*Proposition* #8).

4.3 Behavioral qualities contributing to the success of new practices ("digital transformation leadership qualities")

When analyzing digital transformation practices with respect to their motivational and behavioral components, the 2 × 2 canvas (*Component* B; Sect. 4.1) served to generalize the observed attitudes and behaviors showing up vividly and consistently in contributing to achieving a job to be done. This generalization was based on our understanding of practices put forward in Sect. 3. While practices are aimed towards a goal, the aspect captured with the concept of a job to be done, they would be incomplete or misrepresented without also capturing "ways of understanding, knowing how, ways of wanting and feeling" that are needed for them to be established (Reckwitz 2002, p. 253). In analyzing practice observations, we aimed to capture and general-

ize these motivational and behavioral factors behind the development of practices in each digital transformation practice area. These generalizations of behaviors and attitudes formulated for a practical purpose and playing a vital role in leadership work required in each practice area we called "digital transformation leadership qualities."

The assignment of these qualities to practice areas is not meant to be discriminative or mutually exclusive. Rather, these qualities aggregate signature behaviors and attitudes contributing to the success of practices in one or another digital transformation practice area. The signature behaviors and attitudes indicate that they are strongly expressed and functionally central in each digital transformation practice area – even though individuals working in pursuit of progress in other practice areas may display them to a lesser extent. In other words, the development of practices in a digital transformation practice area benefits from individuals displaying signature behaviors and attitudes for that practice area. Additionally, by being aware of which behaviors and attitudes are signature for the four digital transformation practice areas, organizations can focus on creating a fitting context for these behaviors and attitudes to be displayed, supported, and encouraged.

The identified signature behaviors and attitudes have been summarized into digital transformation leadership qualities, presented in Tables 3, 4, 5 and 6. In a nutshell, as presented to practitioners and used in practice, practices in the vigilant digital transformation practice area are more easily established with (1) a sense of curiosity, (2) openness to alternatives to the current focus, and (3) unbiased thinking (Table 3). The establishment of new ways of working in the voyager digital transformation practice area benefitted from (4) entrepreneurial drive, (5) team co-creation, and (6) team empowerment (Table 4). The development of visionary practices was driven by (7) the power of imagination, (8) future-oriented decision-making courage, and (9) insightful humility (Table 5). New ways of working in the vested digital transformation practice area were established through (10) cross-silo collaboration, (11) lightweight enterprise governance, and (12) meaningful glo-cal change (Table 6).

In sum, the list of jobs to be done (*Component* C; Sect. 4.2), the digital transformation leadership qualities (*Component* D; Sect. 4.3), and the 2 × 2 canvas underlying them (*Component* B; Sect. 4.1), together, constitute the model of digital transformation leadership that sits at the heart of the framework developed in this paper. The model was conceived as an interactive pragmatic tool that is meant to and has been used by participants to collectively design and establish new agile practices that, in combination, digitally transform their organizations. The way we suggest putting the model to work in practice (*Component* E) is discussed in the next section.

5 Putting the digital transformation leadership model to work

Having developed the theoretical account of digital transformation and its leadership (*Component* A; Sect. 3), and presented the model of digital transformation leadership comprising a collection of practical tools (*Components* B, C, & D; Sect. 4), we now move on to discussing a process logic capturing our proposed way of putting the model to work in organizations. The proposed process logic (*Component* E) is grounded in the theoretical propositions and practical experiences with applying

Table 3 Vigilant leadership qualities

Quality	Definition	Examples of signature behaviors and attitudes per area
(1) Sense of curiosity	A desire to explore, learn, and understand more about the world out there.	• Being driven to grasp and respond to the organization's contextual turbulence driven by digital technologies. • Actively scanning the environment for signs of change influenced by digital advancements. • Seeking out diverse cues, signals, ideas, and perspectives through the use of digital tools and platforms. • Continuously learning about technological developments, emerging trends, and shifts in the broader business landscape. • Monitoring opportunities and threats beyond the boundaries of the organization, its business model, or its industry. • Exploring the technological and business context both broadly and deeply with sustained curiosity.
(2) Openness to alternatives to the current focus	A capacity to consider and explore a broad and diverse range of possibilities that reside outside the dominant business perspective.	• Contributing to creating an atmosphere of openness to perspectives that challenge current beliefs. • Making sense of alternative viewpoints and connecting them meaningfully to existing organizational thinking. • Examining current organizational biases and assumptions to better understand and anchor new ideas. • Tethering alternative views to existing decision-making and action frameworks to make them accessible to others. • Collaboratively crafting alternative perspectives to build organizational readiness for diverse future scenarios.
(3) Unbiased thinking	A quality of engaging in reasoning free from personal prejudices, emotions, and preconceived notions.	• Taking a broad and long-term view of potential contextual developments in the digital age. • Engaging with uncertainties, weak signals, trends, and emerging ideas through abductive reasoning to explore plausible scenarios. • Enabling others to logically consider alternative ways the environment and future of work might evolve. • Creating a judgment-free space that encourages critical discussion of multiple believable interpretations of change in organizations' environments. • Making beliefs, assumptions, biases, and preferences explicit to examine their implications – including scenarios where they may not hold true.

the model in organizations. It consists of three steps: (1) laying the groundwork for establishing digital transformation leadership (Sect. 5.1), (2) coordinating the experimental development of agile practices in communities of practice (Sect. 5.2), and (3) resolving the emerging tensions by orchestrating the efforts across the communities (Sect. 5.3).

5.1 Groundwork: socializing digital transformation leadership

When putting the model to work, the goal is to set up systems for and initiate continuous practice change in organizations. To lay the groundwork for using the model of digital transformation leadership, we propose that organizations start by socializing the model among all potential participants. That is because to start changing practices, individuals need to develop a sufficient shared understanding of the common endeavor to serve as a springboard for action (*Proposition* #4). Therefore, organizations need to start by developing a shared understanding of the digitally disruptive environment, digital transformation, and the roles of participants in it. While this

Table 4 Voyager leadership qualities

Quality	Definition	Example leadership behaviors and attitudes per area
(4) Entrepreneurial drive	A quality of coming up with resourceful ways to make innovative ideas tangible.	• Developing resourceful ways to make innovative, digitally-informed ideas tangible and actionable. • Evaluating the business potential of new ideas by demonstrating their value-in-use and emphasizing early, frequent learning. • Taking initiative and focusing energy on navigating the challenges of starting and growing new opportunities, even under turbulent or adverse conditions. • Leveraging the ongoing stream of digital technologies to innovate in more pragmatic, faster, cheaper, and more effective ways.
(5) Team co-creation	A team's capacity to leverage diverse skills and perspectives to generate multi-fold innovative solutions efficiently.	• Nurturing collective creativity within teams by blending diverse perspectives and creative thinking. • Shaping a team's ability to address challenges creatively by promoting inclusion, open debate, and productive conflict grounded in trust. • Instilling a problem-solving approach that values iteration, rapid adaptation, and learning from failure. • Building high-performance teams that embrace and thrive on change.
(6) Team empowerment	A team's capacity to operate in a high-trust environment where stakeholders respect the team's decision-making autonomy.	• Entrusting teams with high degrees of autonomy to decide how to act in response to challenges and opportunities. • Encouraging teams to take responsibility and be accountable for solution-scoping decisions, balancing desirability, feasibility, and viability. • Allowing teams to shape their own work organization and processes to enhance speed, flexibility, and responsiveness. • Creating a high-trust environment where team decisions are respected by organizational stakeholders. • Clearly framing team autonomy within a well-articulated broader organizational value framework to guide empowered decision-making.

Table 5 Visionary leadership qualities

Quality	Definition	Example leadership behaviors and attitudes per area
(7) Power of imagination	A quality of envisioning and expecting to be part of a narrative for a desired possible future of the organization transcending the constraints of the present.	• Imagining and framing the organization's purpose and value-creation model to help it thrive in the digital age. • Envisioning a future-state for the organization through an ecosystem perspective that includes digital win-win partnerships. • Constructing and communicating a compelling and believable narrative of organizational transformation in a turbulent environment. • Inspiring internal and external stakeholders with a clear vision, setting expectations, and building commitment. • Creating direction and momentum to support ongoing and future digital-driven organizational changes.
(8) Future-oriented decision-making courage	A capacity for making bold strategic choices that prioritize long-term benefits and sustainability despite paradoxes, uncertainty, risk, and possible short-term setbacks.	• Making bold, strategic choices by thinking in terms of informed bets and real options for future growth. • Activating new growth paths that involve both radical evolution of the existing core and the creation of adjacent or self-disruptive opportunities. • Addressing decision paradoxes head-on, especially those involving the interplay between people and technology. • Embracing a growth mindset for digital transformation by investing in learning and development for people. • Framing transformation as a positive, emotional, and empathetic journey grounded in purpose. • Instilling digital transformation efforts with a strong sense of meaning and long-term vision.
(9) Insightful humility	A quality of recognizing one's limitations and the value of others' perspectives and feedback, combined with the wisdom to learn and grow from that understanding.	• Understanding the digital landscape and anticipating the organization's future needs with a mindset of humility. • Acknowledging personal limitations and staying open to continuous learning. • Seeking out and embracing evidence and honest feedback from diverse perspectives. • Remaining willing and able to adapt strategies in response to an evolving digital environment. • Balancing strong advocacy for purpose, vision, and transformation principles with inclusive, collaborative decision-making. • Building credibility through a combination of clear direction and openness to emergent insights.

shared reference frame given to an organization's digital transformation endeavor might differ across organizations, on a minimal level, it should be aligned with all the derived analytical propositions (*Component* A; Sect. 3) to put the model to work as it was intended.

When introducing and socializing the model, organizations are proposed to make use of its components designed to involve all willing participants in the digital transformation endeavor. The 2 × 2 canvas of digital transformation paradoxical tensions and practice areas (*Component* B; Sect. 4.1) provides a broad view of the kinds of work involved in digital transformation. The jobs to be done (*Component* C; Sect. 4.2) provide a lower-level guidance of the progress that needs to be achieved with the practice change and allow participants to focus their leadership efforts. The digital transformation leadership qualities (*Component* D; Sect. 4.3) provide an overview of the signature behaviors and attitudes that can drive digital transformation in different practice areas. They illustrate the behaviors and attitudes that can facilitate the

Table 6 Vested leadership qualities

Quality	Definition	Example leadership behaviors and attitudes per area
(10) Cross-silo collaboration	A capacity to foster an inclusive organization of fluid and productive cross-boundary connections and collaborations among individuals.	• Shifting the agile exploitation frontier of digital transformation by fostering fluid, productive cross-boundary collaboration. • Building a networked organization – internally and across the ecosystem – centered on a shared sense of customer value. • Using digital technologies to empower collaboration and streamline operations. • Removing bureaucratic obstacles and breaking down functional silos to improve responsiveness. • Enhancing information flows to reduce delays in decision-making and execution. • Strengthening collaboration both vertically across hierarchies and horizontally across value and supply chains.
(11) Lightweight enterprise governance	A quality of aligning and coordinating organizational stakeholders that promotes and does not stifle swift decision-making.	• Moving beyond visioning and experimentation by steadily building a more productive, flexible, and responsive organization. • Mapping out an enterprise-wide path to succeed with digital transformation, rather than focusing on isolated initiatives. • Designing and evolving a digital target operating model based on modular, reusable digital components. • Acting as architects of enterprise change with a strong bias for practical, sustained action. • Adopting lightweight frameworks for orchestrating and governing transformation to align objectives and coordinate resources. • Integrating transformation efforts across the enterprise without impeding fast strategic execution.
(12) Meaningful glo-cal change	A quality of ensuring that required changes hold significance and benefit both individuals and the organization.	• Engaging the entire organization in digital transformation by ensuring changes are meaningful and beneficial for both individuals and the organization. • Creating motivating work environments that support learning and adoption of shared digital tools, practices, and methods. • Building a strong sense of community by fostering empowerment, collaboration, shared values, and collective purpose. • Guiding the organization in bridging a global digital transformation vision with local realities and a growth mindset. • Actively removing psychological and material barriers to empower individuals throughout the transformation journey.

practice development process and give participants an idea of which digital transformation practice area they want to contribute to. The qualities are designed to signal to participants where the digital transformation work might give them energy, match their strength and potential, and where they might be ready to be held to account. At the same time, they provide an overview of behaviors and attitudes that need to be supported and developed by creating a fitting environment for them in organizations. The 2 × 2 canvas, jobs to be done, and digital transformation leadership qualities all together serve to give participants a shared reference frame and ways in which they

can extend themselves and their responsibilities beyond their current roles towards shaping how their organization will work in the future.

In line with the understanding of digital transformation as continuous practice change, organizations need to ensure participants' continuous and active engagement in practice development. To this end, organizations issue an open invitation to all willing participants to contribute to the change of practices in one or another digital transformation practice area. Extending this invitation means that participation in the organization's digital transformation is effectively open to all willing participants while ensuring that they have enough capacity to contribute to digital transformation leadership within their chosen practice area. Accepting this open invitation, on the part of participants, means that they choose to step into a leadership role. Consequently, they are expected to commit sufficient effort to the digital transformation journey, taking responsibility and accountability for their chosen practice area.

5.2 Experimentation: coordinating agile practices in communities of practice

Having chosen to step into a leadership role (contribute to digital transformation leadership) in one of the practice areas, participants work towards achieving jobs to be done in their chosen area with practices that they develop. This coordinated development of new practices is the work of digital transformation leadership. The establishment of new practices as part of a digital transformation endeavor is inherently collective (*Proposition* #8). Therefore, to facilitate the process of developing and changing practices in ways that align with our model of digital transformation leadership (*Components* B, C, & D; Sect. 4), we propose to create conditions for collective practice development by encouraging the development of communities of practice (Wenger 1998). Communities of practice, emerging spontaneously or developed by organizations, are formed around a "*joint enterprise* as understood and continually renegotiated by its members," characterized by "the relationships of *mutual engagement*," and resulting in a "*shared repertoire* of communal resources […] that members have developed over time" (Wenger 1998, p. 2). Working collaboratively within the resulting communities means "participating in and contributing to the development of practices established by [a] community" (Cobb and Bowers 1999, p. 5). Therefore, to stimulate the collective development of digitally agile practices, organizations encourage the formation of communities of practice around each digital transformation practice area.

When establishing communities of practice around each digital transformation practice area, organizations are proposed to make use of the model components that have been designed to facilitate the collaborative development of new agile practices. Digital transformation practice areas allow for the creation of a minimal shared identity within each community. This shared identity serves to facilitate participants connecting beyond their organizational positions, working collaboratively on developing practices that address the same or similar jobs to be done, and role-modeling complementary leadership qualities. By working on developing practices in resulting communities, participants, on the one hand, are at the forefront of actively developing new practices, and, on the other, they bring their own experience, expertise, and individual cultural and work environment into them. In resulting communities of

practice, participants share and value their collective competence in a given domain, build relationships in which they interact and learn together, and develop a shared repertoire of resources including "experiences, stories, tools, [and] ways of addressing recurring problems" (Wenger-Trayner and Wenger-Trayner 2015).

With communities of leadership practice driving the development of new agile practices, an organization's digital transformation again depends on individuals' active participation. It is up to organizations to create conditions for all participants to be actively involved in the creation of practices and the "continuous renegotiation" of the joint enterprise so that emerging practices are inclusive of different cultural and work groups within an organization. To ensure that the development of practices remains continuous while practices are experimented with, corrected, shared, and adopted, we propose that each community establish regular check-in moments. During these moments, participants would discuss what has been done (by whom and where), what is underway, and what has been proposed. From that picture, potential community experiments are identified and framed as inquiry. Each experiment states a hypothesis about digital-agile progress for the community's leadership agenda, names the change to try, who will be involved, the signals that would show progress, and when to pause and share results. Ideally, the community operates on a predictable learning cycle, for example, by reviewing practice experiments quarterly to synthesize learning and direct attention. That way, the community can come together to guide further practice development before initiating the next experimentation cycle.

5.3 Tension resolution: orchestrating the efforts of communities of practice

With the four leadership communities experimentally developing new agile practices and extending their organization's competence in each of the four practice areas over time, organizations need to ensure that all practice areas are given their due attention. As over time different practice areas can be prioritized at the detriment of others, organizations must establish regular moments of overviewing all agile practices experimented with and resolving possible tensions among them. To that end, the four leadership communities need to regularly come together for discussion and tension resolution.

The model has been designed to be embedded in the logic of paradox management. Defining the four practice areas served the purpose of separating conflicting imperatives to give each their due recognition (*Propositions* #5, 6, & 7), in line with the paradox management logic. However, organizations need to also connect and engage the opposing practice areas (*Proposition* #7), for "actively managing connections between them in order to leverage interdependencies and benefit from their synergies" (Smith et al. 2016).

While within practice areas participants regularly engage in coordinated action to keep track of and share developing practices, the same process needs to be encouraged across practice areas. To ensure that the emerging practices in each practice area compose a coherent digital transformation journey, organizations establish regular check-in moments across the four communities. During these "global" check-in moments, representatives of the four communities of practice come together to have an overview of all emerging practices, evaluate the digital transformation progress

of the whole organization, and resolve any undesirable tensions between emerging practices in different practice areas. Coming together, the discussion across all four leadership communities should be guided by a question of how, in developing practices, each community can help, rather than hurt, the work of other communities.

For example, despite vigilant and voyager practice areas together serving the purpose of exploring new digital opportunities, the voyager practice area can remain more tied to the current ways of working in an organization. That means that the action-driven exploration of new opportunities, often constituted as experimentation in agile teams, might be less bold than the belief-driven exploration of new opportunities in an organization. This temporary mismatch creates a tension between the two practice areas that, if not created consciously for a certain purpose, does not allow the two communities to support each other's exploration. This paradoxical tension can be revealed and discussed during a "global" check-in moment, with a potential solution being to introduce a vigilant practice into the voyager community to stimulate the thinking that is really out of the box in that community.

Another common paradoxical tension may arise between the vested and the voyager practice areas. With both areas focusing on "on the ground" change in ways of working, the voyager community seeks to explore new opportunities, while the vested community aims to exploit known opportunities. Smooth change in practices from action-driven exploration of new opportunities toward integrating them into the organization's exploitation capacity requires both practice areas to work together and support each other's efforts. If the voyager and vested communities are too disconnected from each other's work, there is a tension preventing the smooth transition of practices from the voyager to the vested practice area. This tension can be similarly revealed in a "global" check-in moment. A possible solution can introduce more return-on-investment thinking into the voyager community. This intervention, however, will consciously shift the balance in an organization towards exploitation, which the organization needs to be aware of and correct for, when necessary.

Thus, orchestration is the regular mechanism that brings the poles together, makes tensions discussable, and sends communities back with feedback on their digital-agile practice experiments. Having regular moments of internal evaluation within each community aligned in time and followed by a meeting of all four communities – 4V orchestration – can ensure that the whole organization goes through cycles of learning and improvement of its ways of working at the same cadence. For example, quarterly leadership community experiment reviews can be synchronized with a half-yearly 4V orchestration forum. The evaluation moments within and across communities, then, can infuse the next period of experimentation within the whole organization.

The proposed process logic, therefore, is meant to guide organizations through setting up for and initiating continuous practice change. Within the described logic, the model is socialized organization-wide, allowing for creating a shared reference frame. Consequently, all organization members receive an explicit and open invitation to participate in digital transformation leadership, or in developing, coordinating, and orchestrating changing practices organization-wide. With the help of the 2×2 canvas, jobs to be done, and digital transformation leadership qualities, organization members can imagine whether and in which area they can contribute to

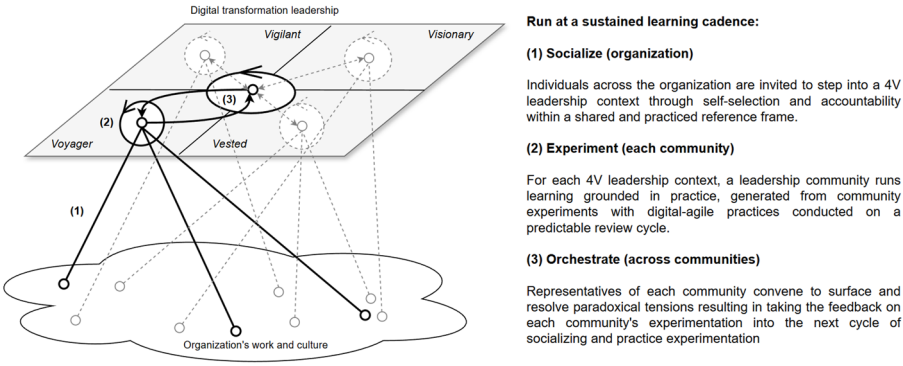

Fig. 4 Meaningful enactment of digital transformation leadership

the work of digital transformation leadership. All willing participants self-select into holding themselves responsible and accountable for the development of practices in their chosen digital transformation practice area. Having chosen a practice area to contribute to, participants experiment to develop practices achieving one or several of the corresponding jobs to be done collectively, in communities of practice. To keep themselves accountable, share the learnings and practices, as well as resolve potential tensions, all communities establish check-in moments within and across communities at a regular cadence. This way, they can take the learnings from their sharing sessions into account for the next experimentation cycle. Figure 4 provides an overview of this process. The figure has been used in organizations to inform the process of putting the model to work.

Figure 4 provides an overview representation of putting the model to work:

The continuous performance and maintenance of the steps represented in Fig. 4 constitute digital transformation leadership driving an organization toward digital-age organizational agility through digital transformation.

6 Discussion

This article aimed to address the following research question: *How can organizations achieve digital-age organizational agility by developing digital transformation leadership?* In response, it put forward a pragmatic framework grounded in understanding digital transformation as a process of collective change and presenting practical tools for facilitating it in organizations. The framework, grounded in theory and informed by practice, consists of an account of digital transformation leadership (*Component* A; Sect. 3), a model of digital transformation leadership synthesized from it (*Components* B, C, & D; Sect. 4), and a proposed process logic explaining how to use the model for guiding the development of digital transformation leadership in organizations (*Component* E; Sect. 5). The framework, in its entirety, is meant to be used as a *poietic* or *generative* organizational artifact that, with sufficient socialization, can enable practitioners' "capacity to make" with respect to their digital transformation.

6.1 Contributions to theory

The framework presented in this paper builds on existing digital transformation literature in connecting the concepts of digital transformation, organizational agility, and leadership (Ly 2023). By following the "practice turn" in research on organizations (Reckwitz 2002; Schatzki et al. 2001; Whittington 2011), similar to strategy-as-practice and leadership-as-practice literature (Carroll et al. 2008; Golsorkhi et al. 2010; Raelin 2016, 2020, 2023), this framework develops propositions calling for approaching digital transformation from the practice lens. That means that the digital transformation endeavor, the concept of organizational agility for digital transformation, and the leadership to drive the transformation have been understood and presented in terms of the change to established practices within organizations. While the connection between the digital transformation agenda and the establishment of new work practices is being explored in recent years (Ashrafi et al. 2025), up till now, to our knowledge, no one has approached the systemic digital transformation effort as a collective process of changing practices in an organization. The implications of this perspective have been formulated into analytical propositions, and while the practical usefulness of the framework provides some validation of them, additional research to verify and explore the propositions is welcome and needed.

In addition, the framework itself, its form, and the process of its development serves as a research contribution. While initiated as a design research project, approaching digital transformation as an organizational endeavor based on practitioner demands did not fit the scope of a singular or a collection of action design research (ADR) projects (Sein et al. 2011). Breaking apart the development of the project into singular design projects would not have allowed approaching digital transformation at different levels of organizing at once, looking at the level of organizational paradoxical tensions (Danneels and Viaene 2022; Soh et al. 2023), communities of practice (Wenger 1998; Wenger-Trayner and Wenger-Trayner 2015), and the importance of individual engagement in the process of collective change through change in practices (Caldwell 2012; Reckwitz 2002; Schatzki et al. 2001). Therefore, while each component of the framework underwent iterative cycles of building, intervention, and evaluation, the framework as a whole co-evolved over more than a decade through a process of pragmatic inquiry. As a result, it serves as a design artifact aimed at addressing a truly practical problem, bridging the gap between pragmatic researchers and reflective practitioners by empowering the latter to design their own solutions to local challenges within the context of digital transformation.

6.2 Contributions to practice

While the framework connects a lot of dots theoretically which is not frequently done, the practical application of the framework is where these connections come to life. The framework has been developed based on a practitioner request and each step of its development has been tested and corrected in practice, repeatedly made sense of by practitioners, and simplified over time to make it more accessible and more actionable. The framework takes participants of digital transformation by the hand in (1) developing a common reference frame, (2) identifying higher and lower-level

digital transformation leadership goals, (3) finding which behaviors and attitudes they want to embody and support in others while establishing new practices, (4) developing a shared identity to figure out new ways of working collectively, which is a prerequisite for them to be adopted and established, all the while (5) ensuring that prioritizing one or another digital transformation practice area at each point in time is a conscious decision rather than a bias. As a result, digital transformation, when understood for what it is – a continuous and collective organizational change – can emerge as a change that is driven and carried by individuals who embody new and changing ways of working. This understanding of digital transformation was co-designed in close collaboration with reflective practitioners and refined through practical application in organizational settings between 2014 and 2024, supporting efforts toward meaningful and people-driven digital transformation.

6.3 Limitations and future directions

This framework takes digital transformation to be a systemic effort emerging from a change in practices and provides guidance for leading digital transformation understood this way. This conceptualization spans several levels of organizing and opens more doors than the presented framework alone can address. From the theoretical side, as mentioned above (Sect. 6.1), the presented analytical propositions (*Component* A; Sect. 3) stem from the current literature on digital transformation but also from the proposed understanding of digital transformation as a continuous and collectively enacted change of practices. The analytical propositions have proven their pragmatic value in developing and using the framework in organizations. With them being theoretically derived and practically useful, there is further need and an opportunity for future research to investigate them empirically, not least because the progress of digital technology continues to race onwards.

From the practical side, while being rooted in a new understanding of digital transformation, the framework in its current state only provides organizations with high-level guidance on where to focus their energy in digital transformation and with minimal process requirements to drive the change. This opens up interesting avenues for future research. For example, researchers could explore more closely how digital transformation practices take shape and what organizations can do to guide them in a desired direction. More specifically, future research could focus on providing organizations with actionable guidance on how to create the right conditions (1) for individuals to actively engage in developing new practices while demonstrating desirable attitudes and behaviors; (2) for collective efforts to be effectively shared and regularly scaled; and (3) for creatively addressing emerging tensions in digital transformation.

Methodologically, the pragmatic framework presented in this article resulted from a multi-project research program that culminated in its abductive construction. Tackling the relatively novel and complex phenomenon of digital transformation required breaking through the boundaries of existing knowledge, which in turn demanded longitudinal, cross-project inquiry (Paradice et al. 2019). Such challenges, as others have noted, call for innovative research approaches (Satwekar et al. 2024). This creates opportunities for further research; for example, extending established methodologies for managerial artefact design – such as design science research (DSR) (Hevner et al.

2004) and action design research (ADR) (Sein et al. 2011) – to better address digital transformation challenges as specified by *Component* A of the framework. Building on this study, future work could include a reflexive meta-analysis of the framework development process or similar research programs. Such an analysis would provide a foundation for advancing methodological guidance on how established approaches such as DSR and ADR can be applied to pragmatically study digital transformation of organizations.

7 Conclusion

This article puts forward a framework of digital transformation leadership. The framework comprises three parts: a novel account of digital transformation leadership summarized in analytical background propositions, a model of digital transformation leadership constituted by a collection of practical tools for organizations to establish and guide the systems of digital transformation leadership, and a process logic providing recommendations for how to use the model in practice. The framework has been developed in cycles of theorizing and practical application over the period 2014–2024. The novelty of the framework lies in its approach to digital transformation, which draws not only on current research in the field but also on a practice theory perspective (*Proposition* #3). As a result, digital transformation is understood as a process of collective and continuous organizational change toward organizational agility in the digital age. From this understanding of digital transformation, the framework develops a concept of digital transformation leadership as the meaningful enactment of systems of continuous practice change. This understanding of digital transformation and digital transformation leadership grounds a model of digital transformation leadership meant to be used by organizations to stimulate and promote this process of continuous practice change, complemented by a process logic explaining how the model can be used most optimally. By studying and calling for empowering the work of emergent leadership that is necessary for digital transformation and that blurs the lines of departments and job roles, the framework adds to the leadership-as-practice agenda, extending it to the research area of digital transformation. This approach to digital transformation comprises a shift in leadership theory and practice. It recognizes digital transformation as a collaborative endeavor that opens the door to and relies on all willing participants working together on changing practices while drawing on their diverse organizational cultural and work contexts.

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Data availability Due to confidentiality, data used for the analysis reported in this manuscript cannot be made available.

Declarations

Competing interests The authors declare no competing interests.

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